Business Requirements Document (BRD)

ProjectTitle:AI-PoweredFAQChatbot(RAG)forMPOnlineServices

1. ExecutiveSummary

The AI-Powered FAQ Chatbot is an intelligent system designed to provide accurate and instantresponsestouserqueriesrelatedtoMPOnlineservices.ItusesRetrieval-Augmented Generation (RAG) to retrieve relevant data from documents and generate context-aware answers. The solution aims to reduce manual support workload, improve user experience, and provide a scalable support system with integrated grievance handling and analytics.

2. BusinessObjectives

- Provide instant and accurate answers to user queries
- Reduce dependency on manual customer support
- Improve accessibility of MP Online service information
- Enable efficient grievance and feed back management
- Generate action able insights through analytics

3. Stakeholders

- **EndUsers:** Citizens using MP Online services
- **SystemAdministrator:** Manages documents ,users ,and system operations
- **SupportExperts:** Handlee scalated queries
- **Organization:** MP Online / Service Provider

4. Project Scope

In Scope:

- AI-based chatbot for answering FAQs
- Document upload and processing(PDF/Image)
- OCR-based text extraction
- Searchable knowledgebase
- Expertes calation system("AskanExpert")
- Grievanceregistrationandtrackingsystem
- Feedbackcollectionsystem
- Analyticsdashboard

OutofScope:

- Voicecall-basedsupport
- Offline functionality
- IntegrationwithexternalgovernmentAPIs(futureenhancement)

5. BusinessRequirements

ChatbotFunctionality

- Thesystemshallallowuserstoaskquestionsthroughachatinterface
- Thechatbotshallprovideanswersbasedonuploadeddocuments
- Thechatbotshallmaintainconversationhistory

KnowledgeBaseManagement

- Thesystemshallallowadministratorstouploaddocuments (PDF/Image)
- ThesystemshallextracttextusingOCR
- Thesystemshallindexandstoreinformationforefficientretrieval

ExpertEscalation

- Thesystemshallprovideanoptiontoescalatequeriestoahumanexpert
- Thesystemshallnotifyexpertsofpendingqueries
- Thesystemshallallowtrackingof responses

Grievance Management

- Usersshallbeabletosubmitcomplaints
- ThesystemshallgenerateauniqueticketID
- Usersshallbeabletotrackcomplaintstatus

FeedbackSystem

- Usersshallbeabletorateresponses
- Thesystemshallstorefeedbackfor analysis

AnalyticsDashboard

- Thesystemshalldisplayfrequentlyaskedquestions
- Thesystemshallidentifyunansweredqueries
- Thesystemshalltrackuseractivityandfeedbacktrends

6. Non-FunctionalRequirements

- **Performance:**Responsetimeshouldbelessthan3seconds
- **Scalability:**Systemshouldsupportincreasingusersanddata
- **Security:**Securehandlingofuserdataandqueries
- **Usability:**User-friendlyinterfaceforbothusersandadmins
- **Availability:**Highsystemuptimeandreliability

7. BusinessProcess Flow

UserQuery→Retrieval→AIResponse→Display→Escalation→ Feedback

1. Usersubmitsaqueryviachatbot
2. Systemretrievesrelevantinformationfromknowledgebase
3. AIgeneratesaresponse
4. Responseisdisplayedtotheuser
5. Ifunresolved→escalatetoexpert
6. Userprovidesfeedback

8. TechnologyConsiderations

- **Backend:**Python(FastAPI)
- **Frontend:**React.js
- **AIIntegration:**OpenAIAPI/GeminiAPI
- **OCR:**Tesseract
- **Database:**PostgreSQL/MongoDB

9. Success Criteria

- Atleast80%ofqueriesresolvedbychatbot
- Averageresponsetimebelow3seconds
- Usersatisfactionratingabove4/5
- Reductioninmanualsupportworkload

10. RisksandAssumptions

Risks:

- OCRinaccuraciesforscanneddocuments
- AI-generatedincorrectresponses(hallucination)
- Handlinglargedocumentdatasets

Assumptions:

- Usershaveinternetaccess
- Documentsprovidedareaccurateandup-to-date
- Expertsareavailableforescalation

11. Future Enhancements

- Multi-languagesupport(Hindi+English)
- Voice-basedinteraction
- Mobileapplicationintegration
- Notificationsystemforupdates
- Self-learningAIimprovements

12. Timeline

Phase	Duration
RequirementAnalysis	1 Week
Development	2–3Weeks
Testing	1 Week
Deployment	1 Week